Name- Kanishk Patel
Class- Btech CSE (D)
Rollno- 26
University Rollno- 2115000505

Assignment-2

Q1 what is distributed database what is difference between distributed database and multiprocessor systems

Ans A distributed database is a database that consists of two or more files located in different sites either on the same network or on entirely different networks.

Difference between distributed database and multiprocessor system-

A multiprocessor is a computer system with two or more central processing units (CPUs) share full access to a common RAM. In shared memory multiprocessors, all the CPUs shares the common memory but in a distributed memory multiprocessor every CPU has its own private memory.

Based on	Multiprocessor	Distributed
Scalability	Harder	Easier
Programmability	Easier	Harder
Degree of transparency	High	Very High
Architecture	Homogenous	Heterogenous
Basic of communication	Shared memory	Message passing

Q4 what do you mean by transparency in distributed database? Also discuss about its types?

Transparency in DDBMS refers to the transparent distribution of information to the user from the system. It helps in hiding the information that is to be implemented by the user.

In distributed Database Management System, there are four types of transparencies.

1. Transaction transparency - This transparency makes sure that all the transactions that are distributed preserve distributed database ~~on~~ integrity and regularity.
2. Performance transparency - This transparency requires a DDBMS to work in a way that if it is a centralized database management system. Also the system should not undergo any downs in performance as its architecture is distributed.
3. DBMS transparency - This transparency is only applicable to heterogeneous types of DDMS as it hides the facts that the local DBMS may be different. This transparency is one of the most complicated transparencies to make use of as a generalization.
4. Distribution transparency - Distributed transparency helps the user to recognize the database as a single thing or a logical entity and if a DBMS displays distribution data transparency then the user does not need to know that the data is fragmented.

Q2 What are the main reasons for and potential advantages of distributed databases? What are the types of distributed database systems? Explain in brief.

Ans Distributed database management has been proposed for various reasons ranging from organizational decentralization and economical processing to greater autonomy. We highlight some of these advantages here.

- Improved performance
- Easier expansion
- Increased reliability and availability
- Management of distributed data with different levels of transparency.

Types of Distributed Database System.

Distributed databases can be broadly classified into homogeneous and heterogeneous distributed database environment each with further sub divisions.

Homogeneous Distributed Databases- In a homogeneous distributed database all the sites use identical DBMS and operating systems.

Types of Homogeneous Distributed Database.

- Autonomous - Each database is independent that functions on its own. They are integrated by a controlling application and use message passing to share data updates.
- Non Autonomous - Data is distributed across the homogeneous nodes and a center or master DBMS co-ordinates data updates across the sites.

Heterogeneous Distributed Databases - In a heterogeneous distributed database, different sites have different operating systems. It is of two types.

- Federated - The heterogeneous database systems are independent in nature and integrated together so that they function as a single database system.
- Un-federated - The database systems employ a central coordinating module through which the databases are accessed.

Q5 What is a fragment of a relation? What are the main types of fragments. Explain in brief? Why is fragmentation a useful concept in distributed database design?

Ans Fragmentation is a process of dividing the whole or full database into various subtables or sub relations so that data can be stored in different systems. The small pieces of sub relations or subtables are called fragments. These fragments are called logical data units and are stored at various sites. It must be made sure that the fragments are such that they can be used to reconstruct the original relation.
It is of three types.

1. Horizontal fragmentation - Horizontal fragmentation refers to the process of dividing a table horizontally by assigning each row or of relation to one or more fragments.

2. Vertical fragmentation - Vertical fragmentation refers to the process of decomposing a table vertically by attributes are columns. In this fragmentation some of the attributes are stored in one system and the rest are stored in other systems.

3. Mixed fragmentation - The combination of vertical fragmentation of a table followed by further horizontal fragmentation of some fragments is called mixed or hybrid fragmentation.

Fragmentation is a database server feature that allows you to control where data is stored at the table level. Fragmentation enables you to define groups of rows or index keys within a table according to some algorithm or scheme.

Q6 Why is data replication useful in DBMS? What typical units of data are replicated?

Ans Replication is useful in improving the availability of data. Depend on the problem you are trying to solve. Data may be replicated row by row, table by table, or database by database, depending on what you need.

If you need realtime databases that are identical with no latency, then you use distributed transactions to post all changes to all servers simultaneously with no delay. If any server can't accept the data the transaction fails and no data is modified. This eliminates latency but introduces reliability problems as any server being down or any physical connection

to any server bring down, causes all transaction to fail.

If you need to be able to operate independently, then you have a number of replication models to choose from depending on how much latency is acceptable and how much autonomy is required.

Microsoft servers have a replication model that allows complete autonomy, but could ~~no~~ theoretically break atomicity.

Q8 Compare the two-tier and three tier client server architectures.

Ans

Two-Tier Database Architecture	Three Tier Database Architecture
1. It is a client server architecture. In two tier the application logic is either buried inside the user	It is web based application. In three tier the application logic or process ~~resides in the middle tier~~ resides in the middle tier it is separated from the data and the user interface.
2. It is easy to build and maintain.	It is complex to build and maintain
3. It ~~is~~ runs slower	It ~~is~~ runs faster
4. It is less secured as client can communicate with database directly	It is secured as client is not allowed to communicate with database directly.
5. It result in performance loss whenever the users increase rapidly	~~It~~ results in performance loss whenever the system is run on Internet but gives more performance than two tier architecture

Q.23 Discuss what is meant by the following terms degree of homogeneity of a DBMS, degree of local autonomy of a DBMS and federated DBMS?

Ans If Degree of homogeneity of a DBMS → all servers and users use identical software to manage the database then that is known as degree of homogeneity of a DBMS.

Degree of local autonomy of a DBMS- If access is permitted to the local server without any restriction then that is known as the degree of local autonomy of a DBMS.

Federated DBMS- federated dbms is the type of meta database management system and it means that multiple autonomous database management system has been mapped into one single unit.

Q7 What is meant by data allocation in distributed database design? what typical units of data are distributed over sites?

Ans The main objective of a data allocation in distributed database is to place the data fragments at different sites in such a way so that the total data transfer cost can be minimized while executing a set of queries.